

scMut cluster size estimation

Qun Li

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1 How to generate data

run 1_estimateClustersize.sh

2 Data source

SomaMutDB 2.0: A comprehensive database for exploring somatic mutations and their functional impact in normal human tissues Dataset1 A body map of somatic mutagenesis in morphologically normal human tissues Dataset2 Single-cell whole-genome sequencing reveals the functional landscape of somatic mutations in B lymphocytes across the human lifespan Dataset3 Somatic genomic changes in single Alzheimer’s disease neurons

2.1 Env settings

```
rm(list = ls())
setwd("/Users/liqun/Desktop/scMutrace-tutorial/ReproducibilityAnalysis/Figure1/")
```

3 libraries

```
library(ggplot2)
library(data.table)
```

```
meta_somatic <- readRDS("../Data/randomCluster/meta_somatic.rds")
df_somatic <- readRDS("../Data/randomCluster/df_somatic.rds")

#saveRDS(meta_somatic, "../Data/randomCluster/meta_somatic.rds")
#saveRDS(df_somatic, "../Data/randomCluster/df_somatic.rds")

colnames(df_somatic) <- c("sampleID", "dist")
df_somatic_meta <- merge(df_somatic, meta_somatic, by = "sampleID")
df_somatic_meta_over10 <- df_somatic_meta[df_somatic_meta$dist >= 10,]
length(df_somatic_meta_over10$sample)/length(df_somatic_meta$sample)
```

```
## [1] 0.9930469
```

```
ecdf(df_somatic_meta$dist)(10)
```

```
## [1] 0.007213078
```

```
df_WGS <- readRDS("../Data/randomCluster/df_WGS.rds")
#saveRDS(df_WGS, "../Data/randomCluster/df_WGS.rds")
colnames(df_WGS) <- c("sample", "dist")
df_WGS_over10 <- df_WGS[df_WGS$dist >= 10,]
length(df_WGS_over10$sample)/length(df_WGS$sample)
```

```
## [1] 0.9458976
```

```
ecdf(df_WGS$dist)(10)
```

```
## [1] 0.05771089
```

4 plot

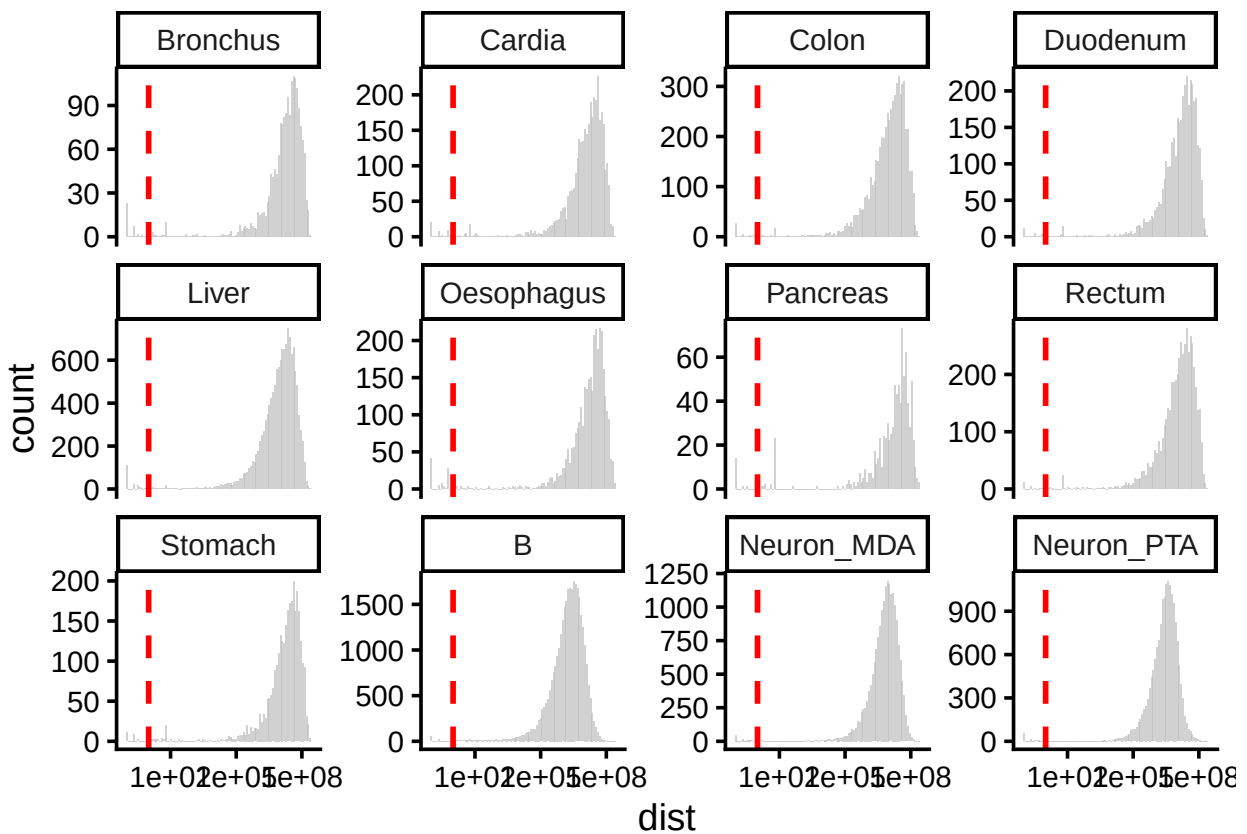
```
threshold_somatic <- quantile(df_somatic_meta$dist, 0.005, na.rm = TRUE)
ecdf_somatic <- ecdf(df_somatic_meta$dist)
quantile_10 <- ecdf_somatic(10)

desired_order <- c("Bronchus", "Cardia", "Colon", "Duodenum", "Liver",
                  "Oesophagus", "Pancreas", "Rectum", "Stomach", "B",
                  "Neuron_MDA", "Neuron_PTA")

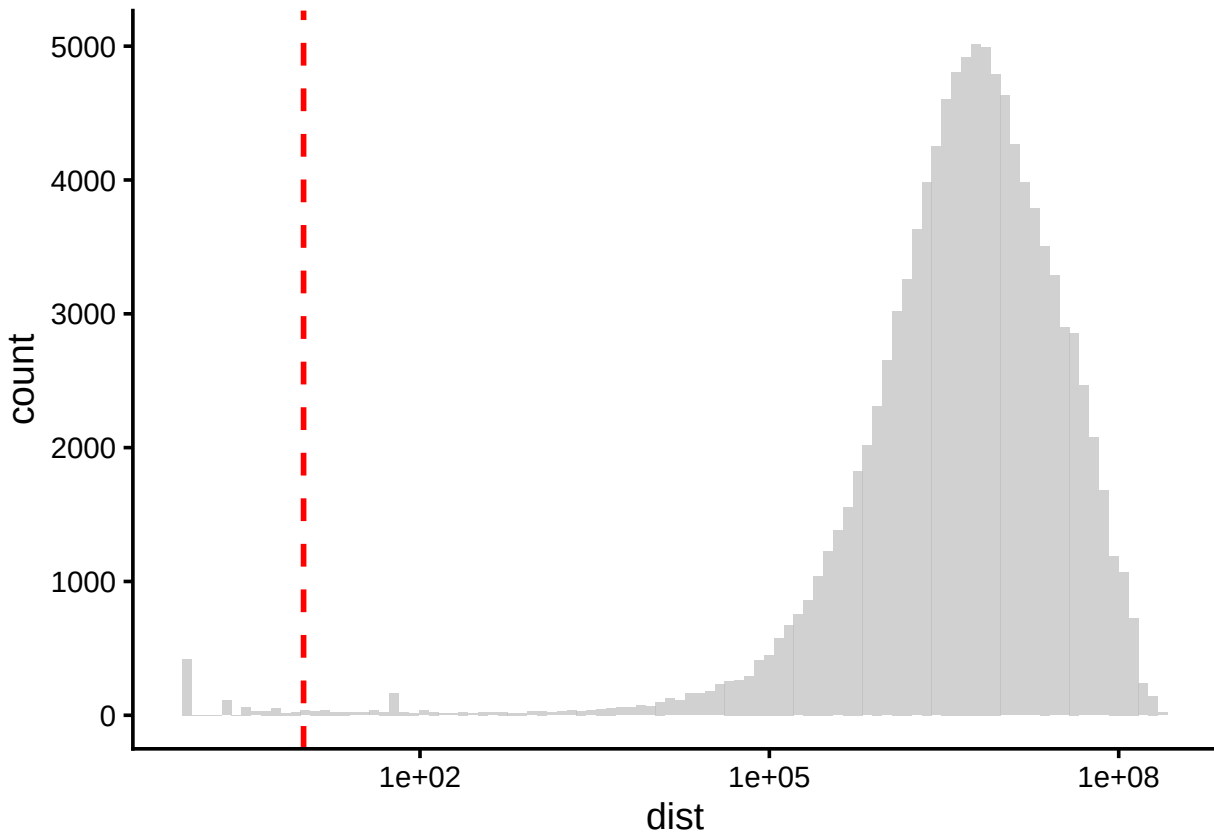
df_somatic_meta$Tissue <- factor(df_somatic_meta$Tissue, levels = desired_order)
```

```
ggplot(df_somatic_meta, aes(x = dist)) +
  geom_histogram(bins = 100, fill = "grey", alpha = 0.7) +
  scale_x_log10() +
  geom_vline(xintercept = 10,
            linetype = "dashed",
            color = "red",
            size = 1) +
  facet_wrap(~Tissue, scales = "free_y") +
  theme_classic(base_size = 14)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



```
ggplot(df_somatic_meta, aes(x = dist)) +
  geom_histogram(bins = 100, fill = "grey", alpha = 0.7) +
  scale_x_log10() +
  geom_vline(xintercept = 10,
            linetype = "dashed",
            color = "red",
            size = 1) +
  theme_classic(base_size = 14)
```



```
my_colors <- c("#1f78b4", "#33a02c", "#e31a1c", "#ff7f00",
               "#6a3d9a", "#b15928", "#a6cee3", "#b2df8a",
               "#fb9a99", "#fdbf6f", "#cab2d6", "#ffff99")

p_somatic_numDis <- ggplot(df_somatic_meta, aes(x = log10(dist), color = Tissue)) +
  geom_density(aes(alpha = 0.3, size = 1, y = ..count..)) +
  #geom_density(aes(alpha = 0.4)) +
  geom_vline(xintercept = 1,
             linetype = "dashed",
             color = "red",
             size = 1) +
  theme_classic(base_size = 14) +
  xlab("Mutation distance (bp, log scale)") +
  ylab("Density") +
  ggtitle("Somatic mutation distance distribution by Tissue") +
  scale_color_manual(values = my_colors)

threshold_wgs <- quantile(df_WGS$dist, 0.005, na.rm = TRUE)

threshold_wgs <- quantile(df_WGS$dist, 0.005, na.rm = TRUE)
ecdf_wgs <- ecdf(df_WGS$dist)
quantile_10 <- ecdf_wgs(10)

ggplot(df_WGS, aes(x = log10(dist))) +
  geom_histogram(bins = 100, fill = "grey", alpha = 0.7) +
  geom_vline(xintercept = 1,
```

```

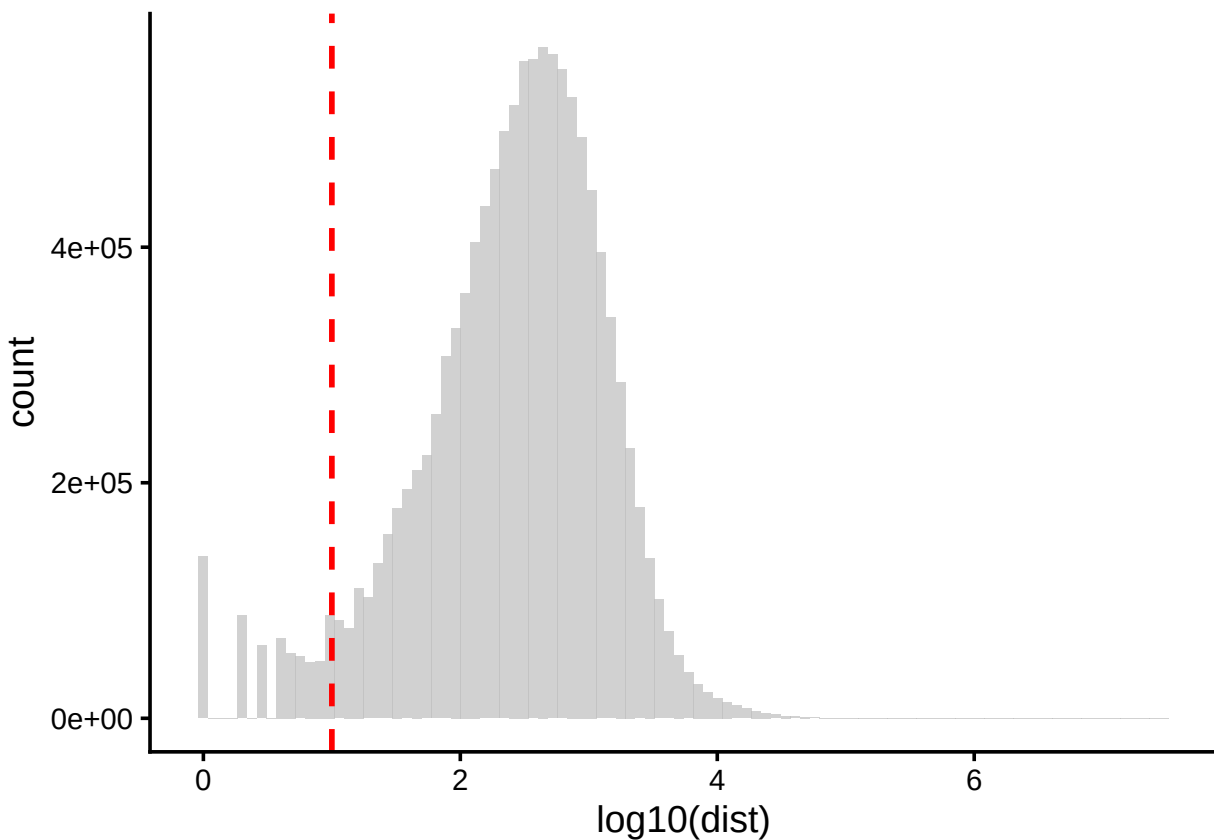
    linetype = "dashed",
    color = "red",
    size = 1) +
  theme_classic(base_size = 14)

```

```

## Warning: Removed 43976 rows containing non-finite outside the scale range
## ('stat_bin()').

```



```

p_WGS_numDis <- ggplot(df_WGS, aes(x = log10(dist), color = sample)) +
  geom_density(aes(alpha = 0.3, size = 1, y = ..count..)) +
  geom_vline(xintercept = 1,
    linetype = "dashed",
    color = "red",
    size = 1) +
  theme_classic(base_size = 14) +
  xlab("Mutation distance (bp, log scale)") +
  ylab("Density") +
  ggtitle("Somatic mutation distance distribution by Tissue") +
  scale_color_manual(values = my_colors)

p_somatic_numDis

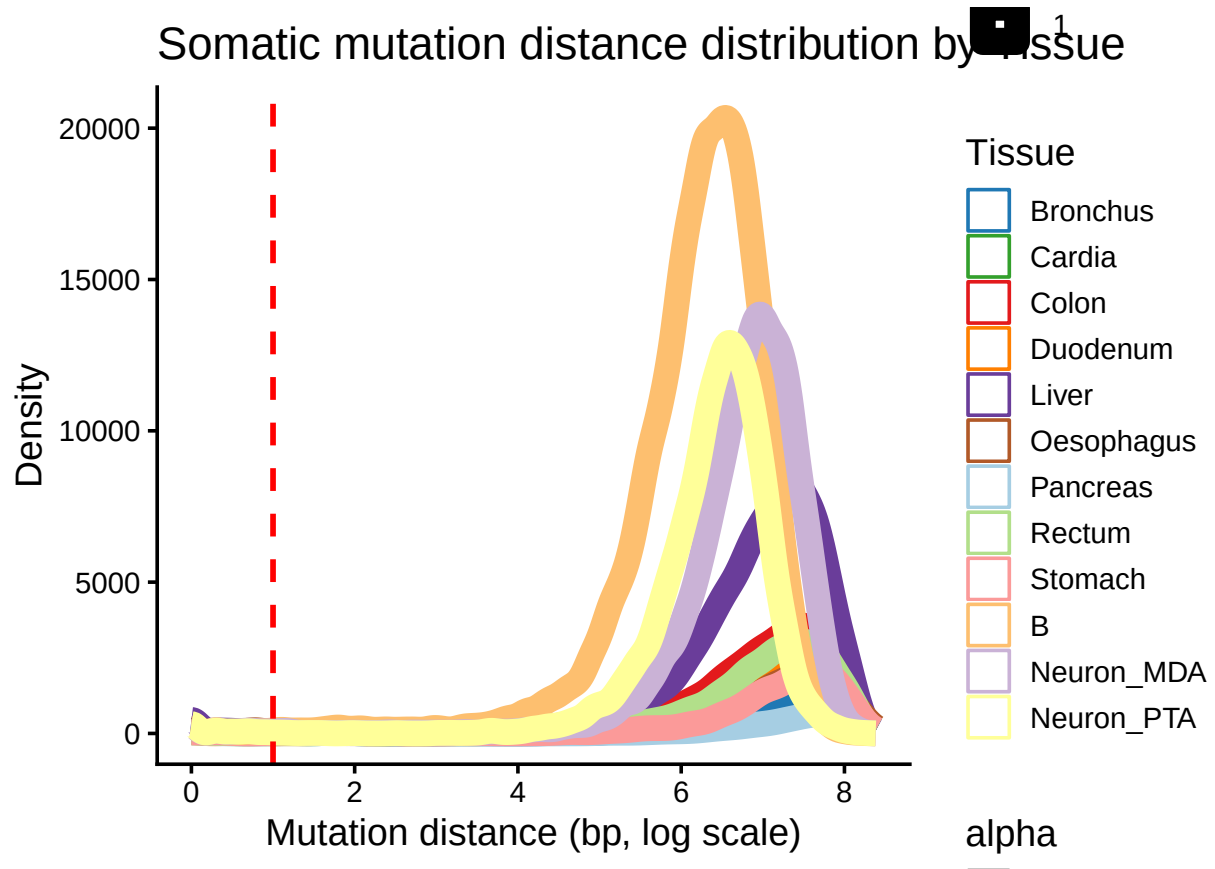
```

```

## Warning: The dot-dot notation ('..count..') was deprecated in ggplot2 3.4.0.
## i Please use 'after_stat(count)' instead.

```

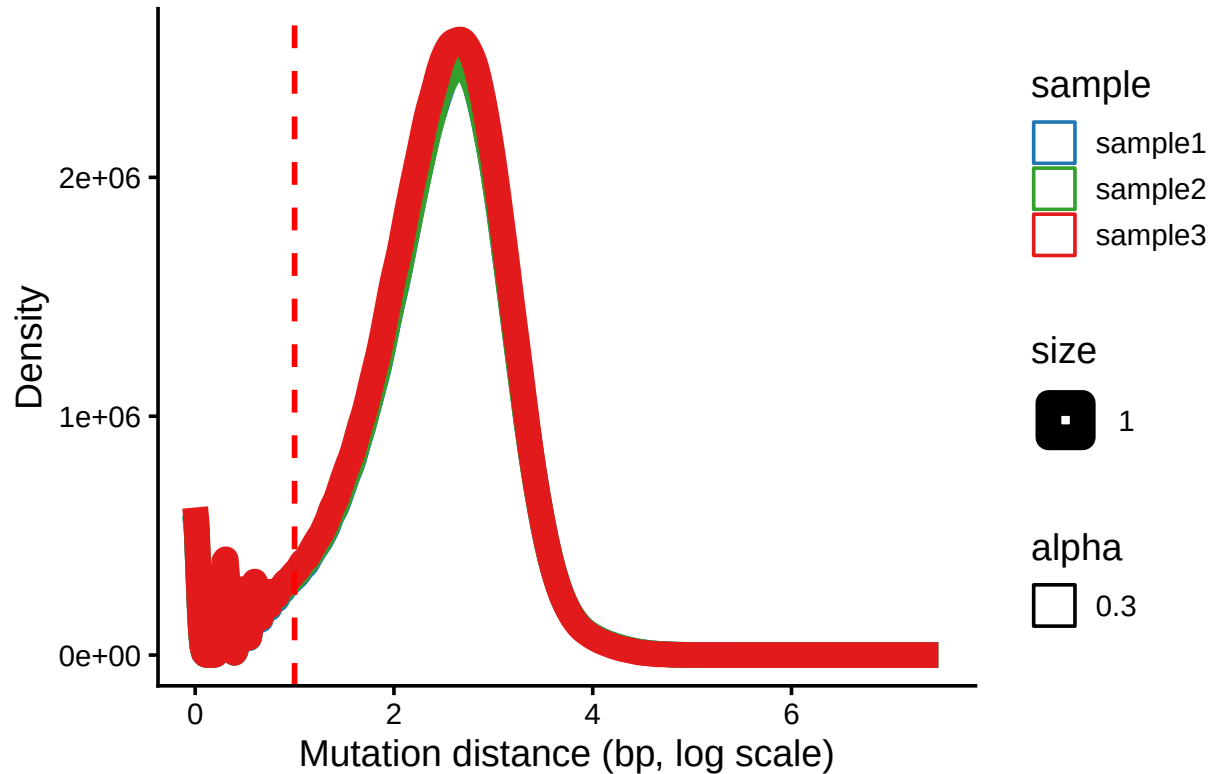
```
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



p_WGS_numDis

```
## Warning: Removed 43976 rows containing non-finite outside the scale range
## ('stat_density()').
```

Somatic mutation distance distribution by Tissue



```
#ggsave("p_somatic_numDis.pdf", p_somatic_numDis, width = 7.21, height = 3.67)
#ggsave("p_wgs_numDis.pdf", p_WGS_numDis, width = 7.21, height = 3.67)
```

5 bin size

```
df_somatic_pos <- readRDS("../Data/randomCluster/df_somatic_pos.rds")
df_wgs_pos <- readRDS("../Data/randomCluster/df_wgs_pos.rds")

#saveRDS(df_somatic_pos, "../Data/randomCluster/df_somatic_pos.rds")
#saveRDS(df_wgs_pos, "../Data/randomCluster/df_wgs_pos.rds")

df_somatic_pos[, bin10 := floor(pos/10)*10]
df_wgs_pos[, bin10 := floor(pos/10)*10]

window_counts <- df_somatic_pos[, .N, by=.(sampleID, chr, bin10)]
window_counts_wgs <- df_wgs_pos[, .N, by=.(sampleID, chr, bin10)]

mutation_dist <- as.data.table(table(window_counts$N))
setnames(mutation_dist, c("N", "N_windows"))
mutation_dist[, N := as.integer(as.character(N))]

mutation_dist$mutation_number_cat <- ifelse(mutation_dist$N >= 3, ">=3", as.character(mutation_dist$N))
mutation_dist <- aggregate(N_windows ~ mutation_number_cat, data = mutation_dist, sum)
```

```

mutation_dist$mutation_number_cat <- factor(mutation_dist$mutation_number_cat, levels = c("0", "1", "2", "3", "4", "5", "6", "7", "8", "9"))

mutation_dist_wgs <- as.data.table(table(window_counts_wgs$N))
setnames(mutation_dist_wgs, c("N", "N_windows"))
mutation_dist_wgs[, N := as.integer(as.character(N))]

mutation_dist_wgs$mutation_number_cat <- ifelse(mutation_dist_wgs$N >= 3, ">=3", as.character(mutation_dist_wgs$N))
mutation_dist_wgs <- aggregate(N_windows ~ mutation_number_cat, data = mutation_dist_wgs, sum)
mutation_dist_wgs$mutation_number_cat <- factor(mutation_dist_wgs$mutation_number_cat, levels = c("0", "1", "2", "3", "4", "5", "6", "7", "8", "9"))

p_somatic_num10bp <- ggplot(mutation_dist, aes(x=mutation_number_cat, y=N_windows)) +
  geom_bar(stat="identity", fill="grey") +
  theme_classic() +
  xlab("Number of mutations per 10bp window") +
  ylab("Number of windows") +
  ggtitle("10bp window mutation count distribution")

p_wgs_num10bp <- ggplot(mutation_dist_wgs, aes(x=mutation_number_cat, y=N_windows)) +
  geom_bar(stat="identity", fill="grey") +
  theme_classic() +
  xlab("Number of mutations per 10bp window") +
  ylab("Number of windows") +
  ggtitle("10bp window mutation count distribution")

table(window_counts$N)

```

```

##
##      1      2      3      4
## 139715  582    7    3

```

```
table(window_counts_wgs$N)
```

```

##
##      1      2      3      4      5      6      7      8
## 11217620 308876 35139 4878  960  207   53  18
##      9
##      4

```

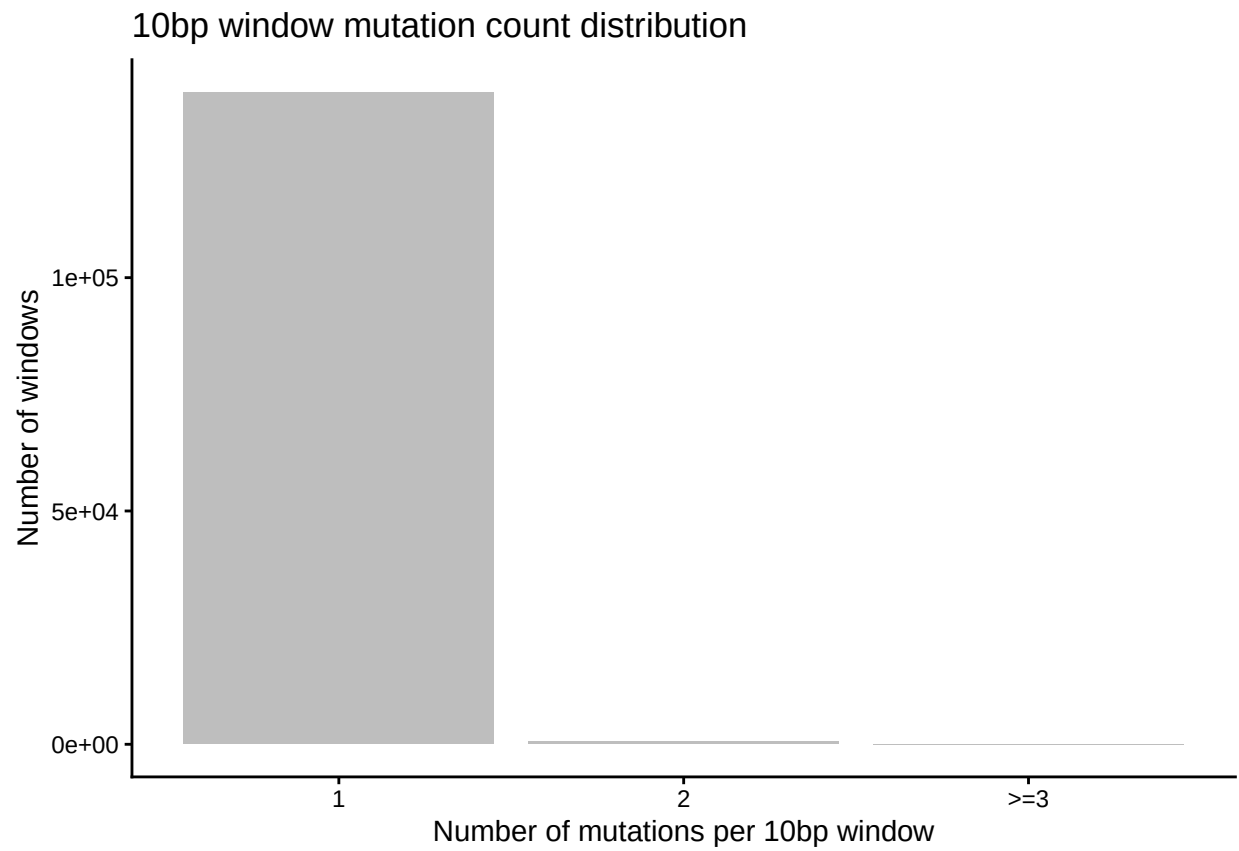
```
ecdf(window_counts$N)(2)
```

```
## [1] 0.9999287
```

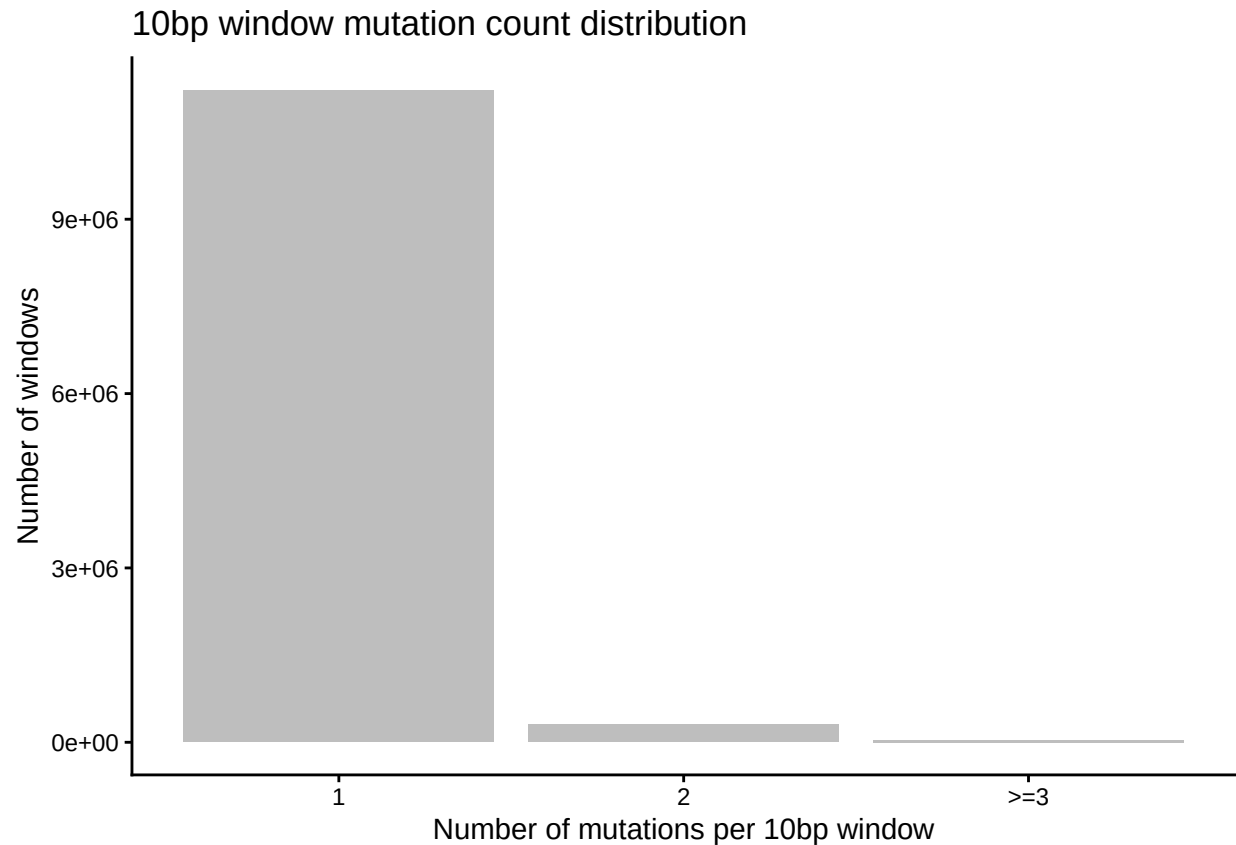
```
ecdf(window_counts_wgs$N)(2)
```

```
## [1] 0.9964333
```

```
p_somatic_num10bp
```

p_wgs_num10bp



```
#ggsave("p_somatic_num10bp.pdf", p_somatic_num10bp, width = 7.21, height = 3.67)
#ggsave("p_wgs_num10bp.pdf", p_wgs_num10bp, width = 7.21, height = 3.67)
```

6 r estimate

```
n <- 10
p <- 0.0075
k <- 0:n
prob <- dbinom(k, size = n, prob = p)

df_prob <- data.frame(
  mutation_number = k,
  Prob = prob
)

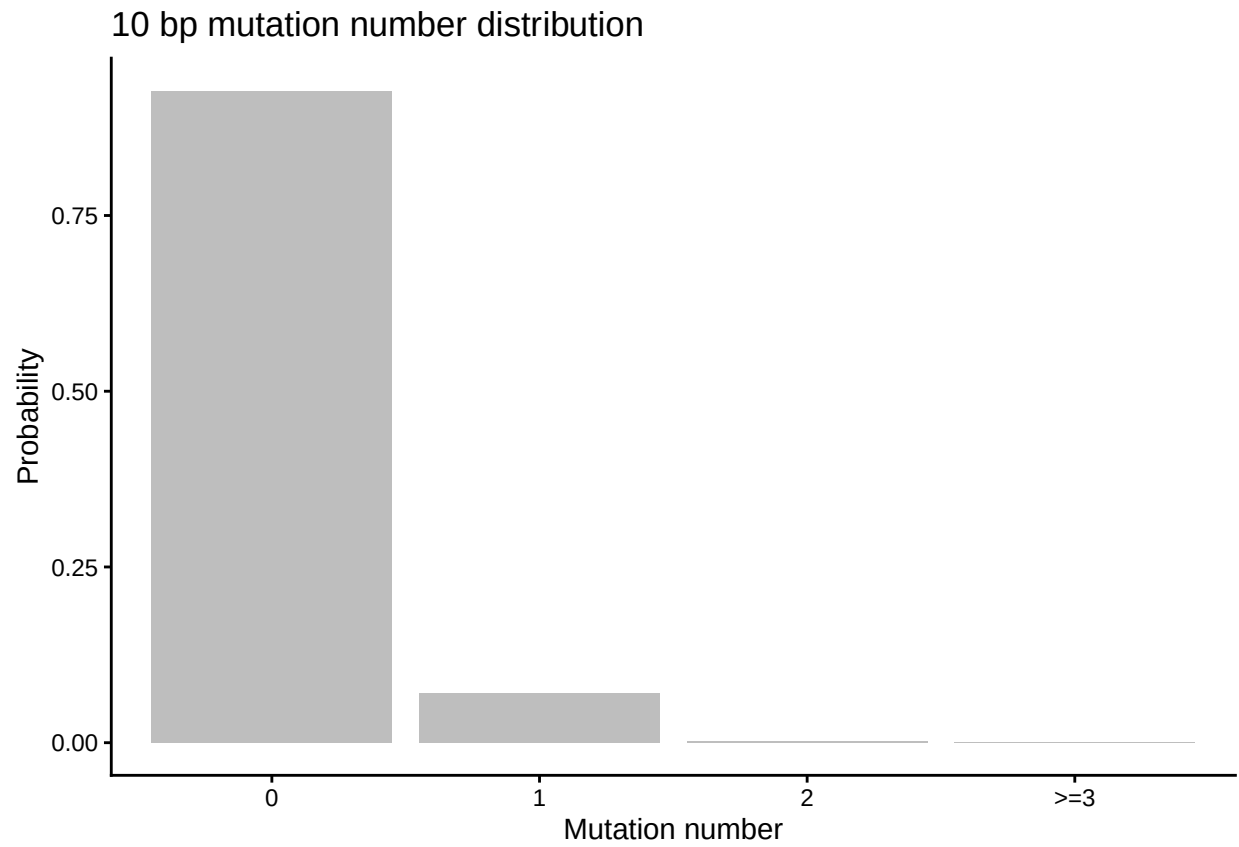
df_prob$mutation_number_cat <- ifelse(df_prob$mutation_number >= 3, ">=3", as.character(df_prob$mutation_number))
df_prob_plot <- aggregate(Prob ~ mutation_number_cat, data = df_prob, sum)

df_prob_plot$mutation_number_cat <- factor(df_prob_plot$mutation_number_cat, levels = c("0", "1", "2", ">=3"))

p_prob <- ggplot(df_prob_plot, aes(x = factor(mutation_number_cat), y = Prob)) +
  geom_bar(stat = "identity", fill = "grey") +
  labs(x = "Mutation number", y = "Probability",
```

```
title = "10 bp mutation number distribution") +  
theme_classic()
```

p_prob



```
#ggsave("p_prob.pdf", p_prob, width = 7.21, height = 3.67)
```